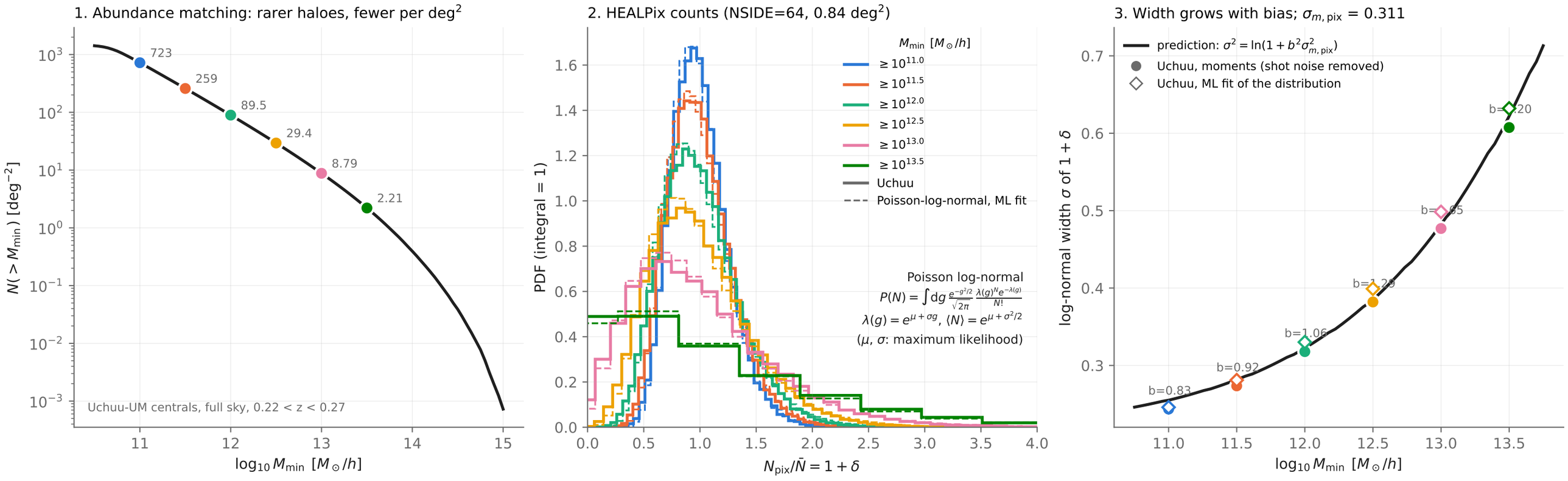
1. Abundance matching: rarer haloes, fewer per deg^2
$N(>M_{\rm min})$ [deg^{-2}]
$\log_{10} M_{\rm min}$ $[M_\odot/h]$
723
259
89.5
29.4
8.79
2.21
Uchuu-UM centrals, full sky, 0.22 < z < 0.27
2. HEALPix counts (NSIDE=64, 0.84 deg^2)
PDF (integral = 1)
$N_{\rm pix}/\bar{N} = 1+\delta$
$M_{\rm min}$ $[M_\odot/h]$
$\geq 10^{11.0}$
$\geq 10^{11.5}$
$\geq 10^{12.0}$
$\geq 10^{12.5}$
$\geq 10^{13.0}$
$\geq 10^{13.5}$
Uchuu
Poisson-log-normal, ML fit
Poisson log-normal
$P(N) = \int dg \frac{e^{-g^2/2}}{\sqrt{2\pi}} \frac{\lambda(g)^N e^{-\lambda(g)}}{N!}$
$\lambda(g) = e^{\mu+\sigma g}$, $\langle N\rangle = e^{\mu+\sigma^2/2}$
(μ, σ: maximum likelihood)
3. Width grows with bias; $\sigma_{m,\rm pix}$ = 0.311
prediction: $\sigma^2 = \ln(1+b^2\sigma^2_{m,\rm pix})$
Uchuu, moments (shot noise removed)
Uchuu, ML fit of the distribution
log-normal width σ of $1+\delta$
$\log_{10} M_{\rm min}$ $[M_\odot/h]$
b=0.83
b=0.92
b=1.06
b=1.20
b=1.35
b=1.50